

ACM — SSL/TLS Certificates

What is ACM?

AWS Certificate Manager (ACM) is a managed service that handles the entire lifecycle of SSL/TLS certificates for your AWS resources — requesting, validating, deploying, and automatically renewing them before they expire.

Without ACM, you would need to manually purchase certificates from a Certificate Authority, install them on each server, track expiry dates, and renew them every year. ACM eliminates all of that overhead entirely.

Traditional Certificate Process:

Buy cert **from** CA (~\$100/year)
Install manually on each server
Track expiry yourself
Renew every year
Update config on renewal

With ACM:

Free with AWS services
Attach **to** ALB **in** 2 clicks
Auto-renews before expiry
Never expires **if** attached
Zero-downtime renewal

The most important thing to understand: **ACM certificates are completely free** when used with integrated AWS services like Application Load Balancers, CloudFront distributions, and API Gateway. You only pay for the underlying AWS service, never for the certificate itself.

Key Concepts

Term	Plain English Explanation
SSL/TLS	The encryption protocol that makes HTTPS work — scrambles data between browser and server so nobody can intercept it
TLS Termination	The ALB decrypts incoming HTTPS traffic and forwards plain HTTP to your backend instances — instances don't need to handle encryption
End-to-End Encryption	HTTPS all the way from browser → ALB → backend instances — traffic is never decrypted in transit
DNS Validation	Proving domain ownership by adding a specific CNAME record to your DNS — recommended method
Email Validation	Proving domain ownership by clicking a link sent to admin emails like admin@yourdomain.com

Term	Plain English Explanation
Wildcard Certificate	A single certificate that covers a domain and ALL its subdomains (* example.com covers www , api , mail , shop , etc.)
Certificate ARN	The unique identifier AWS assigns to your certificate — used when attaching it to ALB or CloudFront

Why Use ACM?

4 Core Reasons:

1. HTTPS is Required Today

Browsers **show "Not Secure" on HTTP** sites

Google penalizes non-HTTPS sites **in search** rankings

Users don't trust sites without the padlock icon

2. Completely Free

No annual certificate fees

No renewal costs

Only pay for the AWS service it's attached to

3. Auto-Renewal

ACM renews certificates automatically **before** expiry

No reminders, no manual work, no downtime from expired certs

4. Native AWS Integration

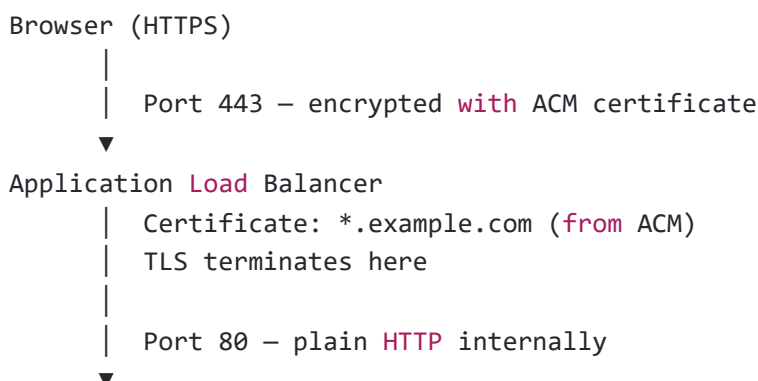
Works directly **with** ALB, CloudFront, API Gateway

Route 53 + ACM = one-click DNS **validation**

No manual installation on servers

Hands-On: Request, Validate, and Attach a Certificate

What We're Building



EC2 Instances / Target **Group**
(backend **never** deals **with** encryption)

Also:

HTTP :80 → ALB → 301 Redirect → **HTTPS** :443
(**users** typing **http**:// automatically **get** upgraded)

Step 1 — Request a Certificate

Navigate to ACM:

- AWS Console → Search "**Certificate Manager**" → Open ACM
- Make sure you are in the **correct region** — certificates are region-specific

 If you are attaching the certificate to CloudFront, you must request it in **us-east-1 (N. Virginia)** regardless of where your other resources are. For ALB, request it in the same region as your ALB.

- Click "**Request a certificate**"
- Select "**Request a public certificate**" → Click "**Next**"

Add Your Domain Names:

Domain	Type	Covers
example.com	Exact match	Root domain only
*.example.com	Wildcard	All subdomains (www, api, shop, mail, etc.)

Best practice: always request both `example.com` AND `*.example.com` together in a single certificate. This covers your root domain and every subdomain with one cert.

Validation Method:

Method	How It Works	When to Use
DNS Validation 	Add a CNAME record to your DNS	Recommended — works with Route 53 automatically
Email Validation	Click link in email to admin@yourdomain.com	Use only if you don't control the DNS

- Select "**DNS validation**" → Click "**Request**"

Step 2 — Validate Domain Ownership

After requesting, ACM gives you a CNAME record that proves you own the domain.

What the validation record looks like:

ACM gives you this CNAME to add to DNS:

Name: `_a79865eb4cd1a6ab56f5b50a..example.com`

Value: `_1234567890abcdef.acm-validations.aws.`

This record proves to AWS that you control `example.com`

Option A — Domain is in Route 53 (Easiest):

- On the certificate details page → expand **Domains** section
- Click "Create records in Route 53" button
- ACM automatically adds the CNAME records to your hosted zone
- Click "Create records" → Done 

Option B — Domain is with GoDaddy or Namecheap:

Go to your registrar's DNS management panel and manually add:

Field	Value
Record Type	CNAME
Name	<code>_a79865eb4cd1a6ab56f5b50a</code> (the part before your domain)
Value	<code>_1234567890abcdef.acm-validations.aws.</code>
TTL	300

Validation Timeline:

Record added → DNS propagates globally → ACM detects it → Certificate issued

Typical wait time: 5 to 30 minutes

Maximum wait time: 72 hours (DNS propagation across globe)

Status changes:

Pending validation → (wait) → Issued 

- Monitor status in **ACM** → **Certificates** — refresh the page every few minutes
- Once status shows "Issued" — your certificate is ready to use

Step 3 — Attach Certificate to ALB

Add HTTPS Listener:

- EC2 → Load Balancers → Select your ALB
- Click the "Listeners and rules" tab
- Click "Add listener"

Field	Value
Protocol	HTTPS
Port	443
Default action	Forward to → select your target group
Default SSL/TLS certificate	Select your ACM certificate from dropdown

- Click "Add" 

Redirect HTTP to HTTPS (Best Practice):

Users might type `http://example.com` directly. You want them automatically upgraded to HTTPS.

- In the Listeners tab → find the **HTTP : 80** listener
- Click on it → **Edit listener**
- Change the default action to: **Redirect to URL**

Field	Value
Protocol	HTTPS
Port	443
Status code	301 — Moved Permanently

- Click "Save changes" 

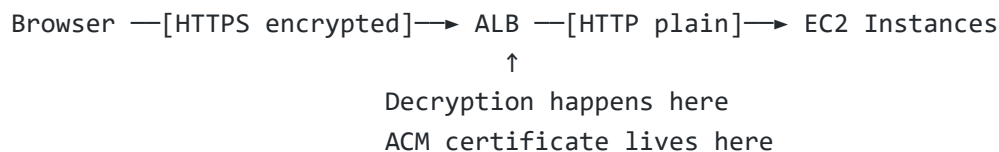
Result of 301 redirect:

User types: `http://example.com`
Browser sends: `GET http://example.com`
ALB responds: `301 → https://example.com`
Browser follows: `GET https://example.com` 
Browser saves: Remembers to always use HTTPS next time

TLS Termination vs End-to-End Encryption

These are two different architectural approaches to where HTTPS encryption lives in your stack.

Option A — TLS Termination at ALB (Most Common)



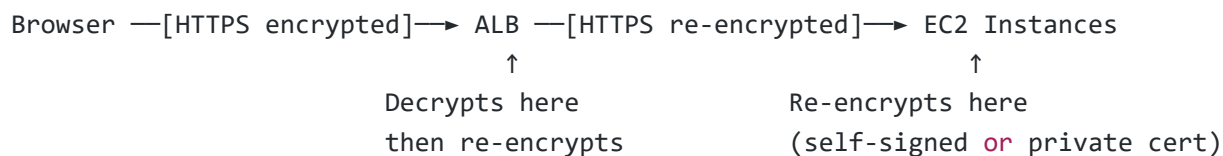
Pros:

- ✓ Simpler backend – EC2 instances don't handle SSL
- ✓ Lower CPU **on** backend instances (no encryption overhead)
- ✓ ACM handles everything – no certs **to** install **on** servers
- ✓ ALB can inspect HTTP headers **for** routing decisions

Cons:

- ⚠ Traffic **between** ALB **and** EC2 **is** unencrypted
- ⚠ (This **is** inside AWS's private network – generally acceptable)

Option B — End-to-End Encryption



Pros:

- ✓ Data encrypted at every hop – even inside AWS network
- ✓ Required **for** strict compliance (PCI-DSS, HIPAA)

Cons:

- ⚠ Must install certificates on every EC2 instance
- ⚠ Higher CPU usage on backend servers
- ⚠ More complex setup **and** maintenance
- ⚠ Target group protocol must be set **to** HTTPS

Which should you use?

Use TLS Termination (Option A) **when**:

- Standard web applications
- Internal backend traffic stays **within** AWS VPC
- You want simplicity **and** **lower** operational overhead

Use End-to-End Encryption (Option B) **when**:

- **Strict** compliance requirements (healthcare, payments)
- Regulatory mandates require encrypted **data at** every layer
- Your **security policy** requires **no** plaintext anywhere

ACM Quick Reference

Region Rules:

ALB certificate → Same region as ALB
CloudFront certificate → MUST be us-east-1 (N. Virginia)

Always Request Both:

example.com → Covers root domain
*.example.com → Covers all subdomains

Validation Methods:

Route 53 + ACM → One-click automatic (recommended)
Other registrar → Manual CNAME record addition

Wait Times:

DNS validation → 5 to 30 minutes typically
Certificate status → Pending validation → Issued

Cost:

Public certificate → Free with ALB, CloudFront, API Gateway
Private certificate → \$400/month per CA

HTTP to HTTPS Redirect:

HTTP :80 listener → Action: Redirect → HTTPS :443 → 301
Always do this – never leave HTTP open without redirect